

BULBUL SAINI

Software Development Engineer (SDE) • Full-Stack & Systems

Ajmer, Rajasthan, India | +91-8112218432 | since2003bulbul@gmail.com | [linkedin.com/in/bulbul-saini-715309261](https://www.linkedin.com/in/bulbul-saini-715309261) | github.com/bulbulsaini

EDUCATION

Women Engineering College — *Bachelor of Technology in Computer Science & Engineering* **Expected May 2026**
Cumulative GPA: 8.02 / 10.0 (Ranked Top 10% in Department) *Ajmer, Rajasthan*

Aryan International Academy — *Senior Secondary (Class XII, RBSE)* — 92.2% Merit **May 2021**

EXPERIENCE

DEE Piping Systems — *Software & Full-Stack Developer Trainee* **Jun 2026 – Present**
Production Software Engineering *India*

- **Selected** to build and maintain full-stack production features using **Java**, **React.js**, and **Node.js**, decreasing client render latency by **28%**.
- **Engineered 8+ RESTful API endpoints** with JSON validation, processing **1,000+ daily requests** at 99.9% uptime.
- **Refactored** core service layers applying architectural design patterns (MVC, Factory), eliminating **25% redundant code** across database access layers.

Rajasthan Urban Infrastructure Development Project (RUIDP) — *Engineering Intern* **Jun 2025 – Aug 2025**
SCADA Infrastructure Monitoring (Asian Development Bank funded) *Abu Road, Rajasthan*

- **Selected** for the prestigious ADB-funded Women in Engineering Fellowship to analyze real-time **SCADA telemetry systems** across municipal networks.
- **Automated** data verification scripts to monitor multi-sensor telemetry feeds, reducing manual verification hours by **30%**; awarded **Grade A** for technical excellence.

Birla Institute of Scientific Research — *Java Development Intern* **May 2024 – Jul 2024**
Applied Research & Software Quality Assurance *Jaipur, Rajasthan*

- **Orchestrated** Java algorithm optimizations using the **Java Collections Framework** (HashMaps, PriorityQueues), reducing search complexity from O(N) to O(1) and accelerating query speed by **35%**.
- **Constructed** automated test verification procedures and executed **50+ manual QA test cases** alongside QA leads, identifying and resolving **15+ defects** prior to launch.

TECHNICAL PROJECTS

Full-Stack Authentication & Authorization Platform — *TypeScript, React.js, Node.js, SQL* **Jan 2026 – Mar 2026**

- **Architected** an end-to-end authentication system supporting **500+ active user accounts** with registration, session management, and RBAC security.
- **Deployed JWT (JSON Web Tokens)** with HTTP-only cookies and **bcrypt** password hashing (10 salt rounds), securing **100% of sensitive API routes** against XSS/CSRF attacks.
- **Formulated** normalized **MySQL** schemas with indexed primary and foreign keys, decreasing credential lookup query latency by **40%**.

Movie Ticket Booking Simulation System — *Java, Multithreading, MVC Architecture, Data Structures* **Aug 2025 – Oct 2025**

- **Programmed** a multi-screen cinema booking application managing **10+ movie catalogs**, **50+ showtimes**, and concurrent seating grids.
- **Standardized** real-time seat matrix locking using **2D arrays and Java HashMaps**, eliminating **100% of double-booking conflicts** under simulated peak loads.
- **Integrated** structured exception-handling covering **12+ boundary edge cases**, guaranteeing zero runtime crashes during transaction simulations.

Responsive E-Commerce Web Platform (Amazon Clone) — *JavaScript, React.js, CSS Grid, Flexbox* **Feb 2025 – Apr 2025**

- **Developed** a full-featured e-commerce interface supporting **50+ product listings** with dynamic cart updates, category filtering, and checkout flows.
- **Constructed** modular UI components with mobile-first CSS Grid and Flexbox layouts, reducing layout shift by **25%** across all device viewports.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, C, SQL

Web & Backend Frameworks: React.js, Node.js, Express.js, HTML5, CSS3, RESTful APIs, JSON, Responsive Web Design

Databases & Developer Tools: MySQL, Relational Database Design, Database Indexing, Git, GitHub, Docker, Postman, VS Code

Core Computer Science & SDE: Data Structures & Algorithms, Object-Oriented Programming (OOP), Java Collections Framework, DBMS, Operating Systems

HONORS & CERTIFICATIONS

- **Academic & Peer Leadership:** Mentored 15+ junior engineering students in Data Structures, Java OOP, and software design patterns.
- **Asian Development Bank Fellowship:** Grade A Commendation for SCADA Telemetry & Urban Infrastructure Analysis.
- **Java Software Architecture & Algorithms Specialization** — Coursera